



SparseDNA: Efficient Genomic Sequence Modeling via Sparse Experts in State Spaces

Zhehan Xie¹, Ping Han¹, and Chen Lin^{1,2}(✉)

¹ School of Informatics, National Institute for Data Science in Health and Medicine
Xiamen University, Xiang'an South Road, Fujian 361000, China

{xiezhahan, hanping}@stu.xmu.edu.cn

² Zhongguancun Academy, Daniu Fang Second Ring Road,, Haidian District Beijing
100094, China

chenlin@xmu.edu.cn

Abstract. Genomic DNA encodes diverse regulatory mechanisms that control gene expression across different functional and cellular contexts. Existing genomic foundation models often struggle to generalize across tasks because they rely on shared dense parameters to represent highly heterogeneous sequence patterns. To address this limitation, we propose SparseDNA, a hybrid genomic language model that integrates sparse Mixture-of-Experts (MoE) modules into a state-space architecture. By replacing dense projections with sparsely activated expert subnetworks, our model enables adaptive specialization for fine-grained modeling of heterogeneous genomic signals—such as promoters, enhancers, and histone marks—while maintaining computational efficiency. This design significantly increases the effective parameter count and enhances the model’s ability to capture diverse genomic dependencies without substantially raising training costs. Pretrained on the human reference genome at single-nucleotide resolution, SparseDNA achieves state-of-the-art performance on 10 of 18 datasets from the Nucleotide Transformer benchmark, with an average accuracy gain of 1.3% and a notable 2.0% improvement on histone-related classification tasks. Further, the ablation and controlled comparison studies collectively demonstrate that the performance gain is driven by architectural innovation in expert specialization. These results indicate that combining state-space modeling with conditional expert computation provides a scalable and biologically grounded foundation for modeling the regulatory complexity of genomic sequences.

Keywords: Genomic Language Modeling · State Space Models · Mixture of Experts

1 Introduction

Understanding the complex functional patterns encoded in genomic sequences is a fundamental challenge in computational biology. With the exponential growth

Z. Xie and P. Han—Contribute equally to this paper.

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of sequencing data, there is an increasing demand for scalable and efficient models capable of capturing both long-range dependencies across the genome and fine-grained functional semantics at the nucleotide level [1]. Recent advances in large-scale language modeling have inspired a new generation of “genomic foundation models,” such as DNABERT [11, 23] and Nucleotide Transformer [4], which treat DNA as a symbolic language composed of nucleotide tokens.

However, the prevailing Transformer-based architectures face critical limitations when applied to extremely long DNA sequences due to their inherent quadratic computational complexity and constraints on context length [2, 21], hindering their practical application for full genome-scale analysis. To address the inefficiency of the Transformer, recent efforts have explored alternative architectures. The State Space Model (SSM) [2, 10], particularly its selective variant like Mamba [9], has emerged as a promising direction. SSMs overcome the computational bottleneck by achieving linear time complexity for sequence processing, effectively resolving the computational scalability challenge and enabling efficient modeling of dependencies across millions of base pairs.

Despite their efficiency, current dense SSM-based models still face a crucial representational limitation that they share with dense Transformer models: they fundamentally rely on shared dense parameters to represent a wide array of highly heterogeneous sequence patterns—such as promoters, enhancers, and various histone modification marks. These diverse regulatory elements operate under fundamentally different biochemical principles. Consequently, models using dense, shared projections often struggle to achieve the necessary adaptive specialization required for fine-grained modeling, leading to difficulties in generalization across distinct functional and cellular contexts. Effectively capturing this genomic heterogeneity, without sacrificing computational efficiency, remains a critical gap.

To address these dual limitations of efficiency and adaptive specialization, we propose SparseDNA, a novel hybrid genomic language model, which strategically integrates sparse Mixture-of-Experts (MoE) modules into a State-Space architecture (SSM). This design establishes a new paradigm for achieving both linear computational efficiency and specialized, high representational capacity in genome-scale modeling. The core innovation of SparseDNA lies in introducing the principle of “*adaptive specialization*” to SSM-based genomic models. We replace the standard dense feed-forward projections within the State Space Model blocks with sparsely activated expert subnetworks. This conditional computation mechanism enables specialized adaptation for fine-grained modeling of heterogeneous genomic signals. A dynamic router selectively activates a small subset of experts for each genomic token, allowing different experts to specialize in distinct biological functions or sequence motifs. This design significantly increases the effective parameter count and enhances the model’s ability to capture diverse genomic dependencies, all while maintaining the computational efficiency inherent to sparse computation and SSMs, thus avoiding a substantial increase in training or inference costs.

Pretrained on the human reference genome at single-nucleotide resolution, SparseDNA demonstrates superior performance and generalization capabilities. It achieves state-of-the-art performance on 10 of 18 datasets from the comprehensive Nucleotide Transformer benchmark, yielding an average accuracy gain of 1.3%. Notably, SparseDNA shows a remarkable 2.0% improvement on challenging histone-related classification tasks, confirming the efficacy of adaptive specialization for heterogeneous signal modeling. The ablation study shows that moderate expert specialization with multi-expert activation is key to SparseDNA’s performance. The comparative experiment under equal compute confirms that SparseDNA’s improvement comes from specialized experts, not merely increased parameters. These results empirically demonstrate that combining efficient state-space modeling with conditional expert computation provides a powerful and biologically grounded foundation for large-scale genomic sequence modeling.

Our key contributions can be summarized as follows:

1. **The SparseDNA Architecture:** We propose SparseDNA, a novel hybrid genomic foundation model that embeds the *Mixture-of-Experts (MoE)* mechanism within a *State-Space architecture (SSM)* framework, pioneering a solution that achieves both computational efficiency and high representational capacity for genome-scale tasks.
2. **Adaptive Specialization:** We introduce the principle of “adaptive specialization” to SSM-based genomic models. By replacing dense parameters with sparse expert subnetworks, SparseDNA effectively resolves the limitation of shared dense weights struggling with highly heterogeneous genomic signals.
3. **State-of-the-Art Performance:** We establish new state-of-the-art results on the Nucleotide Transformer benchmark, with significant and consistent performance gains, particularly in tasks requiring fine-grained distinction of regulatory elements like histone marks.

2 Preliminaries and Related Work

2.1 Transformer-Based Genomic Language Models

Inspired by the success of Natural Language Processing (NLP), Genomic Language Models (GLMs) treat DNA sequences as a symbolic language composed of nucleotide tokens. Early foundational models such as DNABERT [11] applied self-supervised pretraining objectives to capture functional and structural patterns within genomic sequences, achieving strong performance in tasks such as promoter classification, enhancer prediction, and variant effect estimation. More sophisticated iterations, including the Nucleotide Transformer [4] and various domain-specific pretraining strategies, further refined sequence encoding.

While these Transformer-based models [4, 7, 11, 22–24] have established the state-of-the-art in many fine-grained genomic analyses, they fundamentally rely on the self-attention mechanism, which incurs quadratic computational complexity $O(N^2)$ relative to the sequence length N . This limitation severely constrains

the effective context window, often preventing models from analyzing long-range dependencies across the millions of base pairs necessary for full genome-scale analysis [16]. This inherent inefficiency has spurred the search for alternative, computationally linear architectures.

2.2 State Space Models

To overcome the scalability bottleneck of the Transformer, recent efforts have focused on architectures that achieve efficient long-range sequence modeling [9, 15, 16, 18]. State Space Models (SSMs) have emerged as a powerful paradigm, originating from continuous-time dynamical systems. Unlike earlier recurrent architectures such as RNNs and LSTMs, which struggled with vanishing gradients and limited memory capacity over long ranges, modern SSMs are engineered to integrate information efficiently.

The development of structured SSMs, notably S4 (Structured State Space Sequences) [10], demonstrated that sequence modeling can be performed with linear-time complexity during inference and sub-quadratic complexity during training, making them highly scalable. The critical breakthrough came with Mamba [9], which introduced a selective state-space mechanism. By conditioning the SSM parameters on the input sequence, Mamba effectively filters irrelevant information and retains necessary context, achieving data-dependent information flow while maintaining hardware efficiency and linear scaling. This combination of efficiency $O(N)$ and strong representational capability makes SSMs, and Mamba specifically, ideally suited for handling the vast context lengths characteristic of genomic data.

2.3 Mixture-of-Experts Architectures

Despite the efficiency gains provided by SSMs, current dense models (both Transformer and SSM-based) rely on shared, monolithic parameter sets. Genomic sequences are characterized by high functional heterogeneity; distinct regulatory elements operate under fundamentally different biochemical rules. Modeling this diversity with shared dense projections limits a model’s ability to achieve the necessary adaptive specialization.

Mixture-of-Experts (MoE) architectures offer a robust solution to this representational limitation by enabling conditional computation. The core principle of MoE is to increase model capacity substantially while maintaining constant computational cost per input token. This is achieved by employing a router function that activates only a sparse subset of specialized expert networks for any given input [19].

The concept of MoE has been widely successful in natural language processing (NLP) models [5, 6, 14], with notable examples such as DeepSeekMoE [3] and Mixtral 8x7B [13]. For instance, DeepSeekMoE demonstrated that expert sparsity could lead to highly efficient models. The DeepSeekMoE 16B model, which achieves comparable performance to Llama 2 7B [20] while reducing computational costs by about 40%, is a prime example of this efficiency. Similarly, Mixtral

AI’s Mixtral 8x7B model, built on the Mistral 7B [12] architecture, introduces eight experts per layer, achieving remarkable performance with only a fraction of the total parameters activated, surpassing or matching models like Llama-2-70B [20] and GPT-3.5 [17] in several benchmarks.

In the context of genomics, MoE provides the crucial mechanism for adaptive specialization: distinct experts can implicitly learn specialized representations for different functional classes. By integrating sparse MoE mechanisms with efficient SSM backbones, we aim to develop a model that simultaneously achieves computational linearity and the necessary architectural specialization to capture complex genomic heterogeneity.

3 Method

3.1 Overview of SparseDNA

SparseDNA, as shown in Fig. 1, introduces a fine-grained hybrid architecture that integrates the Mixture-of-Experts (MoE) mechanism directly into the internal computations of the Mamba block.

Instead of interleaving separate MoE layers, we replace the *Gate* and *Convolution* projections in each Mamba block with expert mixtures, enabling conditional computation and adaptive specialization within the state-space dynamics. This design retains Mamba’s linear-time long-range modeling capability while introducing sparse expert activation for enhanced expressiveness. As a result, the model efficiently learns both global and local genomic patterns at single-nucleotide resolution.

3.2 The Mamba Block

The original Mamba architecture models sequences through a selective state-space representation. For an input sequence x_t , the latent state s_t evolves as:

$$s_t = \sigma(W_g x_t) \odot (A s_{t-1} + B x_t), \quad (1)$$

where W_g represents the gating projection, and A, B are state transition parameters. In practice, Mamba introduces additional convolutional mixing over the hidden sequence to capture local context:

$$h_t = \text{Conv1D}(x_t; W_c), \quad (2)$$

where W_c represents the convolutional projection, and then the process is followed by selective updates of the latent state. These Gate and Convolution projections control how information flows across positions and time steps, serving as the key components governing sequence selectivity and locality.

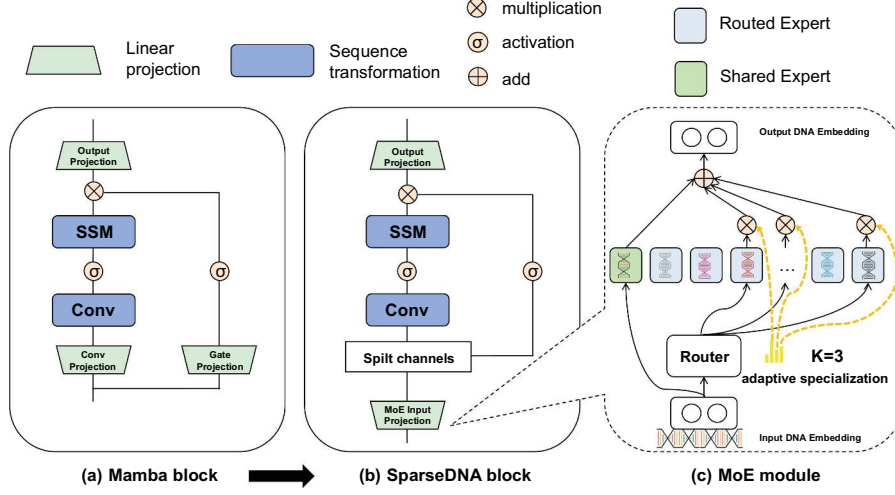


Fig. 1. Overview of the proposed SparseDNA architecture. (a) **Mamba block**: the standard Mamba structure where the input sequence is projected into Convolutional and Gating branches, processed in parallel, and fused by an output projection. (b) **SparseDNA block**: the integration of the MoE projection into the Mamba block, where the MoE replaces the input projection and its output is split into convolution and gating branches. (c) **MoE module**: the Mixture-of-Experts component consists of one shared expert and multiple routed experts. A router performs dynamic sparse activation by selecting the Top-K experts for each DNA segment. Different sequence patterns—illustrated by color-coded DNA fragments—are **adaptively** routed to **specialized** experts that capture diverse regulatory elements (e.g., promoters, enhancers, TF-binding motifs). A shared expert remains active to ensure robust and stable representations. Outputs from the shared and routed experts are fused to produce the transformed features used by subsequent Mamba computations.

3.3 MoE for Gate and Convolution Projections

To enhance stability and generalization, we extend the standard Mixture-of-Experts (MoE) design by introducing a **shared expert** alongside the routed experts. Each projection (Gate and Convolution) is replaced with a shared-expert MoE function:

$$f_{\text{MoE}}(x) = \sum_{k=1}^{N_e} p_k(x) E_k(x) + E_{\text{shared}}(x), \quad (3)$$

where E_k denotes the k -th routed expert, and E_{shared} is a shared expert applied to all inputs. The routing probabilities $p_k(x)$ are generated by a lightweight gating network:

$$p_k(x) = \text{Softmax}(W_r x)_k. \quad (4)$$

During training, a limited number of experts are activated per input (e.g., top- K routing). This shared-expert MoE design provides both conditional special-

ization (through routed experts) and global consistency (through the shared expert), improving the representational diversity and robustness of the Gate and Convolution projections.

Unlike conventional MoE in NLP where each expert is a full feed-forward network (FFN), our design employs **lightweight linear experts**. This significantly reduces computational and parameter overhead. Specifically, each expert is parameterized as a single linear projection:

$$E_k(x) = W_k x + b_k, \quad (5)$$

where $x \in \mathbb{R}^{d_{\text{model}}}$ denotes the input embedding, d_{inner} is the expanded hidden dimension used in the Mamba block, $W_k \in \mathbb{R}^{2d_{\text{inner}} \times d_{\text{model}}}$, and b_k is the bias term.

3.4 SparseDNA Block

Building upon the standard Mamba block (Sect. 3.2) and the shared-expert MoE formulation (Sect. 3.3), we integrate the MoE mechanism into the input projection stage of each Mamba block to form the SparseDNA block.

Given an input hidden sequence $X \in \mathbb{R}^{B \times L \times d_{\text{model}}}$, the original Mamba block first applies a linear projection to produce a joint representation that is later split into Convolutional and Gating branches. In SparseDNA, this projection is replaced by a shared-expert MoE function:

$$X' = f_{\text{MoE}}(X), \quad (6)$$

where $f_{\text{MoE}}(\cdot)$ follows Eq. (3) and aggregates the outputs from routed and shared experts.

The resulting features are then processed identically to the standard Mamba design. Specifically, the MoE output is projected into a joint representation and split into two branches:

$$[X_c, X_g] = \text{Split}(X'), \quad (7)$$

where X_c and X_g denote the Convolutional and Gating inputs, respectively. The Convolution branch captures local contextual interactions, while the Gating branch modulates selective state updates in the SSM:

$$H_c = \text{Conv1D}(X_c), \quad H_g = \sigma(X_g). \quad (8)$$

These components are subsequently used in the selective state-space update and fused via the output projection, preserving the original Mamba computation pipeline.

This design introduces conditional computation only at the projection stage while keeping the SSM dynamics unchanged. By activating a small subset of experts per token, SparseDNA increases representational diversity and specialization with limited computational overhead. At the same time, the always-active shared expert ensures stability and consistent feature transformation across different sequence patterns.

As a result, the SparseDNA block retains Mamba’s linear-time sequence modeling capability while enabling adaptive expert specialization for genomic sequence representation.

4 Experiments

4.1 Pretraining

We pretrain SparseDNA on the human reference genome [8] using a causal language modeling (CLM) objective, predicting the next nucleotide at each position. For input representation, we adopt a single-nucleotide tokenization scheme, treating each nucleotide (“A”, “C”, “G”, “T”, and “N”) as a separate token, along with special tokens for padding, separation, and unknown characters. This approach preserves fine-grained sequence resolution, enabling the model to capture both local motifs and long-range dependencies.

Our model consists of 4-layer decoder-only blocks with a hidden dimension of 256, trained on sequences of length 1024. Each block integrates a Mixture-of-Experts (MoE) module, featuring one shared expert and a set of routed experts, with 1–7 activated routed experts among 31–63 candidates. This design allows us to study the impact of expert sparsity and activation on model performance. The model is trained for 10–20k steps for main evaluations. All models are implemented in PyTorch and trained on NVIDIA A800 GPUs.

4.2 Downstream Tasks

Dataset. We evaluate our model on the Nucleotide Transformer (NT) benchmark, which consists of 18 genomic prediction datasets spanning three major categories: histone marker prediction, regulatory element annotation (including enhancer and promoter classification), and splice site annotation. Each dataset involves DNA sequences ranging from 200 to 600 nucleotides.

Evaluation Metrics. Following the evaluation protocol of [4], we report the Matthews Correlation Coefficient (MCC) for histone marker and enhancer tasks, and the F1-score for promoter and splice site prediction tasks, except for the splice sites *all* task, where we report accuracy.

Baselines. For comparison, we include several strong and widely adopted DNA sequence modeling baselines:

- **HyenaDNA** [16]: a state-space-based long-context genomic model utilizing implicit convolution, enabling in-context learning on genomic tasks.
- **DNABERT-2** [23]: an improved Transformer-based genomic language model employing a BPE tokenizer and whole-genome masking, enabling longer context handling with reduced memory cost.

- **Nucleotide Transformer** [4]: a large-scale DNA foundation models pre-trained on the 1000 Genomes Project and 850 additional species for broad generalization.
- **Caduceus-Ph** [18]: a convolution-state-space hybrid based on Mamba with architectural support for bidirectionality and reverse-complement equivariance.
- **Mamba** [9]: the original selective state-space model serving as our direct architectural baseline.

These baselines cover a range of modeling paradigms, including transformers, convolutional architectures and state-space models, enabling a comprehensive evaluation of SparseDNA across different task types.

Table 1. Nucleotide Transformer benchmark. Performance of HyenaDNA, DNABERT-2, Nucleotide Transformer v2 (NT-v2), Caduceus-Ph, Mamba, and SparseDNA. Results for HyenaDNA, Mamba, and SparseDNA are obtained in this work; results for DNABERT-2, NT-v2, and Caduceus-Ph are taken from [18]. SparseDNA results are obtained using a shared-expert MoE configuration with 31 routed experts and Top-3 activated experts per token (1+3/31), which provides the best performance-efficiency trade-off. We highlight the best values in **bold** type and underline the second best.

Dataset	HyenaDNA 1.6M	DNABERT-2 117M	NT-v2 500M	Caduceus-Ph 1.9M	Mamba 1.8M	SparseDNA 34.5M
<i>Average</i>	0.730	0.679	0.698	0.725	<u>0.765</u>	0.778
<i>Histone Markers Avg.</i>	0.628	0.551	0.572	0.635	<u>0.683</u>	0.705
H3	0.793	0.785	0.784	0.815	0.830	<u>0.817</u>
H3K4me1	0.529	0.511	0.550	0.523	<u>0.600</u>	0.618
H3K4me2	0.470	0.336	0.319	0.487	<u>0.578</u>	0.586
H3K4me3	0.546	0.352	0.410	0.544	<u>0.641</u>	0.708
H3K9ac	0.614	0.542	0.562	0.622	<u>0.642</u>	0.680
H3K14ac	0.629	0.516	0.551	0.631	<u>0.690</u>	0.706
H3K36me3	0.630	0.591	0.625	0.601	<u>0.678</u>	0.691
H3K79me3	0.692	0.613	0.626	0.697	<u>0.729</u>	0.742
H4	0.774	0.796	<u>0.799</u>	0.811	0.785	0.789
H4ac	0.606	0.463	0.495	0.621	<u>0.659</u>	0.716
<i>Regulatory Annotation Avg.</i>	0.795	0.767	0.778	0.775	<u>0.806</u>	0.814
Enhancers	0.571	0.516	0.548	0.546	<u>0.578</u>	0.610
Enhancers Types	0.537	0.423	0.424	0.439	<u>0.561</u>	0.563
Promoter All	0.963	<u>0.971</u>	0.976	0.970	0.966	0.966
Promoter Non-TATA	0.957	<u>0.972</u>	0.976	0.969	0.966	0.964
Promoter TATA	0.948	0.955	0.966	0.953	0.958	<u>0.966</u>
<i>Splice Site Annotation Avg.</i>	0.961	0.959	0.983	0.942	<u>0.963</u>	0.960
All	<u>0.966</u>	0.939	0.983	0.940	0.961	0.959
Acceptors	0.965	<u>0.975</u>	0.981	0.937	0.970	0.966
Donors	0.953	<u>0.963</u>	0.985	0.948	0.958	0.952

Performance on the Nucleotide Transformer Benchmark. To ensure comparability with existing works such as HyenaDNA and Nucleotide Transformer, we attach a lightweight linear decoder to our pretrained backbone and fine-tune it for each downstream task under identical settings.

As shown in Table 1, SparseDNA achieves state-of-the-art performance on 10 out of 18 datasets (Top-1) and ranks within the Top-2 on 12 datasets of the Nucleotide Transformer benchmark.

Compared to its non-MoE counterpart (the Mamba baseline), SparseDNA yields an average improvement of **1.3%** across all tasks and **2.0%** on the subset of ten histone-mark prediction tasks, demonstrating that the introduced MoE mechanism substantially enhances representation capacity without increasing model size or pretraining data requirements.

4.3 Ablation Study on Expert Configuration

To evaluate the contribution of the MoE design, we ablate two key factors in the expert configuration: the number of routed experts (N_e) along with the number of activated experts per token (N_a).

We denote a configuration as $1 + N_a/N_e$, where **1** refers to the shared expert, and N_a/N_e refers to activating N_a routed experts among N_e candidates.

We evaluate the following settings:

$$1 + 1/15, \quad 1 + 1/31, \quad 1 + 3/31, \quad 1 + 7/31, \quad 1 + 7/63$$

Table 2 reports the impact of varying the routed expert pool size (N_e) and the number of activated experts (N_a). Figure 2 further illustrates the empirical trend across expert configurations consistent with the results in 12 NT tasks.

Table 2. Ablation study on expert configuration on the Nucleotide Transformer benchmark. Results are averaged over 12 datasets: 10 histone marker tasks and 2 enhancer regulatory annotation tasks; all metrics are reported as MCC. We selected this subset because they represent the most challenging tracks in the full benchmark, providing a rigorous evaluation of the model’s capability. We denote the MoE setup as $1 + N_a/N_e$, where **1** indicates a shared expert, and N_a/N_e denotes activating N_a routed experts among N_e candidates.

MoE Configuration	MCC
$1 + 1/15$	0.6796
$1 + 1/31$	0.6788
$1 + 3/31$	0.6855
$1 + 7/31$	0.6875
$1 + 7/63$	0.6819

We first analyze the effect of the number of activated experts while fixing the routed expert pool to 31. Using a single activated expert (Top-1 routing, $1+1/31$) yields the poorest results (0.6788), suggesting that limited conditional capacity restricts representational flexibility. Increasing the activation rate to top-3 substantially boosts performance (+0.67% over $1+1/31$), and top-7 activation

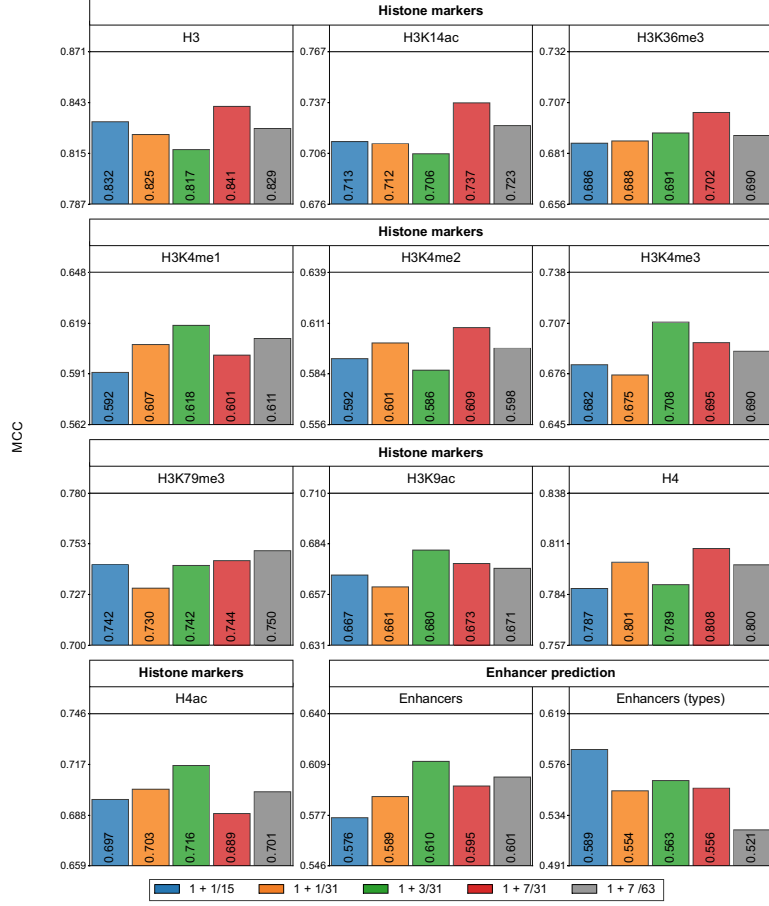


Fig.2. Performance comparison under different expert configurations. Model performance is evaluated on the NT-Histone (10 histone-related tasks) and NT-Enhancer (2 enhancer tasks) datasets using Matthews Correlation Coefficient (MCC). Results demonstrate how varying the number of experts influences model capability in regulatory element prediction under matched compute conditions.

achieves the best score (0.6875), indicating that multiple experts collaboratively contribute to richer context modeling.

Next, we examine the impact of the routed expert pool size. Increasing the number of routed experts does not provide further gains in either setting: expanding the pool from 15 to 31 experts under Top-1 routing (0.6796 vs. 0.6788) or from 31 to 63 experts under Top-7 routing (0.6875 vs. 0.6819) yields comparable or slightly worse performance, likely due to increased routing sparsity and reduced expert utilization.

These results indicate that moderate expert specialization with multi-expert activation is the most effective design for SparseDNA. Considering both per-

Table 3. Comparison of SparseDNA with Dense-MoE and the Mamba baselines on 18 Nucleotide Transformer downstream tasks. Top-1 and Top-2 Wins count the number of datasets where the model ranks first or within the top two, respectively. ‘#’ stands for “number of” In the line of “# Experts”, $a + b$ (e.g., 1+31) denotes a shared experts and b routed experts. In the line of “# Activated Experts”, $a + b$ denotes a activated shared experts and b activated routed experts. ‘# Total Model Params’, ‘# Total Expert Params’ and ‘# Activated Expert Params’ are in millions.

Metric / Model	SparseDNA	Dense-MoE	Mamba
# Experts	1 + 31	4 + 0	1 + 0
# Activated Experts	1 + 3	4 + 0	1 + 0
# Total Model Params (M)	34.5	4.9	1.8
# Total Expert Params (M)	33.9	4.2	1.1
# Activated Expert Params (M)	4.2	4.2	1.1
Avg. Score	0.778	0.767	0.765
Top-1 Wins	12	4	2
Top-2 Wins	13	11	12
H3	0.817	0.840	0.830
H3K4me1	0.618	0.585	0.600
H3K4me2	0.586	0.586	0.599
H3K4me3	0.708	0.668	0.641
H3K9ac	0.680	0.656	0.642
H3K14ac	0.706	0.702	0.690
H3K36me3	0.691	0.665	0.678
H3K79me3	0.742	0.711	0.729
H4	0.789	0.784	0.785
H4ac	0.716	0.691	0.659
Enhancers	0.610	0.601	0.578
Enhancers Types	0.563	0.540	0.561
Promoter All	0.966	0.965	0.966
Promoter Non-TATA	0.964	0.966	0.966
Promoter TATA	0.966	0.959	0.958
Splice Sites All	0.962	0.960	0.961
Splice Sites Acceptors	0.966	0.973	0.970
Splice Sites Donors	0.952	0.961	0.958

formance and efficiency, we adopt Top-3 routing with 31 routed experts as the default configuration in all main experiments.

4.4 Comparison with Dense-Activated MoEs Under Equal Compute

To address the potential confounder of increased model capacity, we construct a Dense-MoE counterpart where all experts are always activated and uniformly fused, removing routing and specialization.

We further scale the Dense-MoE such that its computational cost (FLOPs) matches our sparse $(1 + 3/31)$ configuration. This ensures that both models consume the same amount of computation per token.

Under equal-compute conditions, SparseDNA still achieves superior results on the NT benchmark (as shown in Table 3), demonstrating that:

- Performance gains are **not** merely due to increased parameters;
- Sparse routing enables **conditional computation**, allowing experts to specialize in different sequence patterns;
- Efficient expert activation leads to a better compute-performance trade-off.

5 Conclusion

In this work, we introduce SparseDNA, a novel hybrid genomic foundation model designed to overcome the critical dual challenges facing large-scale genomic sequence modeling: the quadratic computational inefficiency of standard Transformers and the representational limitations imposed by shared dense parameters in existing architectures.

SparseDNA establishes a new architectural paradigm by synergistically integrating the sparse Mixture-of-Experts (MoE) mechanism into a State-Space Model (SSM) backbone. This design maintains the SSM’s linear computational complexity for efficient, long-range context modeling, while simultaneously introducing adaptive specialization through sparse activation. This specialization enables distinct expert subnetworks to learn the unique biological signatures of highly heterogeneous genomic elements, effectively resolving the “compromise representation” problem inherent in dense models. Empirical evaluation on the Nucleotide Transformer benchmark confirmed the effectiveness of our approach.

SparseDNA represents a significant step forward in building truly scalable and functionally specialized Genomic Foundation Models. Future work will focus on extending this framework to enable zero-shot prediction across diverse species and cell types, as well as integrating multi-modal genomic data to further enrich the model’s specialized functional understanding.

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